

Performance Testing

Date	12 July 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing + Flask Test Server
Type of Testing	Functional & Load Testing
Target Module / API	Crop Prediction Module
Test Environment	Local System (Flask) / Render Deployment
Test Date	12-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Crop prediction	1	30 sec	Correct crop predicted
2	Multiple predictions	10	60 sec	No errors
3	Invalid input	5	30 sec	Validation message shown
4	Continuous usage	20	120 sec	Stable performance

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.3 sec	Pass	Fast
2	Response Time (Max)	< 5 seconds	2.6 sec	Pass	Acceptable
3	Throughput (Req/sec)	20 req/sec	24 req/sec	Pass	Good
4	Error Rate	< 1%	0%	Pass	Stable
5	CPU Utilization	< 80%	45%	Pass	Efficient
6	Memory Utilization	< 80%	50%	Pass	Normal

Step 4: Observations & Analysis

- Crop prediction is accurate and fast.
- System performs well under normal load.
- Slight delay with repeated requests.
- Weather API integration not implemented.
- Optimized ML model loading.
- Improved input validation.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):


